

Approach to Hypothesis Testing with Fixed Probability of Type I Error

1. State the null and alternative hypotheses.
2. Choose a fixed significance level α .
3. Choose an appropriate test statistic and establish the critical region based on α .
4. Reject H_0 if the computed test statistic is in the critical region. Otherwise, do not reject.
5. Draw scientific or engineering conclusions.

Significance Testing (P-Value Approach)

1. State null and alternative hypotheses.
2. Choose an appropriate test statistic.
3. Compute the P-value based on the computed value of the test statistic.
4. Use judgment based on the P-value and knowledge of the scientific system.

Variance known

1. $H_0: \mu = \mu_0$
 2. $H_1: \mu \neq \mu_0$
 3. Test statistic: $z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
 4. Reject H_0 if $z > z_{\alpha/2}$ or $z < -z_{\alpha/2}$

1. $H_0: \mu = \mu_0$
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1. $H_0: \mu = \mu_0$
 2. $H_1: \mu < \mu_0$
 3. Test statistic: $z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
 4. Reject H_0 if $z < -z_\alpha$

Test Procedure for a Single Mean (Variance Known)

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} > z_{\alpha/2} \quad \text{or} \quad z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < -z_{\alpha/2}$$

If $-z_{\alpha/2} < z < z_{\alpha/2}$, do not reject H_0 . Rejection of H_0 , of course, implies acceptance of the alternative hypothesis $\mu \neq \mu_0$. With this definition of the critical region, it should be clear that there will be probability α of rejecting H_0 (falling into the critical region) when, indeed, $\mu = \mu_0$.

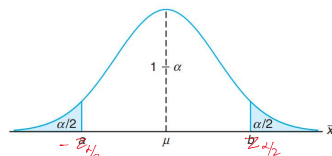
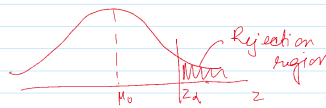


Figure 10.9: Critical region for the alternative hypothesis $\mu \neq \mu_0$.

One-sided Right-tailed

$$H_0: \mu = \mu_0, \\ H_1: \mu > \mu_0.$$

The signal that favors H_1 comes from large values of z . $z > z_\alpha$



Left-tailed one-sided

$$H_0: \mu = \mu_0 \\ H_1: \mu < \mu_0$$

favoring H_1 from small values of z

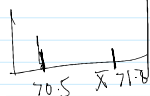


Example 10.3: A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance.

P-value

$$= P(Z > 2.02 | H_0) \\ = 0.0217 \\ < 0.05 (= \alpha)$$

Reject H_0



$$H_0: \mu = 70$$

$$H_1: \mu > 70$$

$$\sigma = 8.9$$

$$\bar{X} = 71.8$$

$$z_{0.05} = 1.65 \\ z_{cal} = 1.64 \\ n = 100$$

$$z_{cal} = \frac{\bar{X} - 70}{\sigma/\sqrt{n}} = \frac{71.8 - 70}{8.9/\sqrt{100}} = \frac{1.8 \times 10}{8.9} = 2.02$$

$$z_{cal} > z_{0.05} \Rightarrow \text{Reject } H_0$$

at 0.05 level of significance, data seems to indicate that mean life span is greater than 70 yrs

$$z_{calc} = 2.02$$

$$P(Z > 2.02) = 0.0217$$

$$H_0: \mu =$$

$$H_1: \mu >$$

$$n = 100, \quad \bar{x} = 71.8, \quad \sigma = 8.9 \\ \alpha = 0.05$$

$$\text{test statistic } z = \frac{\bar{x} - 70}{\sigma/\sqrt{n}} \quad (\text{under } H_0)$$

$$\text{Reject } H_0 \text{ if } z_0 > z_\alpha$$

$$z_0 \text{ is the value of the } z =$$

Reject H_0

$$P\text{-value} = P(Z >) =$$

$$z_{0.05} = 2.575$$

$$z = -2.83$$

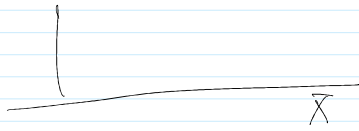
$$P(|z| > 2.83) = 0.0046$$

$$< 0.01$$

Example 10.4:

A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that $\mu = 8$ kilograms against the alternative that $\mu \neq 8$ kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance.

alternative that $\mu \neq 8$ kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance.



$$P(|Z| > 2.58) = 0.0046 < 0.01$$

Reject H_0

$$\begin{aligned} H_0: \mu &= 8 & n &= 50 & \sigma &= 0.5 \\ H_1: \mu &\neq 8 & \bar{x} &= 7.8 \\ Z &= \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} & Z_0 &= \frac{7.8 - 8}{0.5/\sqrt{50}} = -2.8 \\ \text{Reject } H_0 & \text{ if } Z > Z_{\alpha/2} \text{ or } Z < -Z_{\alpha/2} \\ & \text{Reject } H_0 & Z_{\alpha/2} &= 2.575 \end{aligned}$$

$$\left| \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right| > Z_{\alpha/2} \Rightarrow \mu \neq 8$$

Relationship to Confidence Interval [EXACT SAME REASONING]

Reject H_0 if $|Z| > Z_{\alpha/2}$ or if $\mu_0 \notin CI$
Do not reject H_0 if $|Z| < Z_{\alpha/2}$ or if $\mu_0 \in CI$

Therefore, we now know how to do tests for mean $\left\{ \begin{array}{l} \text{known } \sigma^2 \\ \text{unknown } \sigma^2 \end{array} \right.$

pairs of means $\left\{ \begin{array}{l} \text{known } \sigma_1^2, \sigma_2^2 \\ \text{unknown } \sigma_1^2, \sigma_2^2 \end{array} \right.$

$\left\{ \begin{array}{l} \text{unknown but equal} \\ \text{unknown \& unequal} \\ \text{dependent popns} \end{array} \right.$

Table 10.3: Tests Concerning Means

Rejection Region

H_0	Value of Test Statistic	H_1	Critical Region
$\mu = \mu_0$	$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}; \sigma \text{ known}$	$\mu < \mu_0$	$z < -z_\alpha$
		$\mu > \mu_0$	$z > z_\alpha$
		$\mu \neq \mu_0$	$z < -z_{\alpha/2} \text{ or } z > z_{\alpha/2}$
$\mu = \mu_0$	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}; v = n - 1, \sigma \text{ unknown}$	$\mu < \mu_0$	$t < -t_\alpha$
		$\mu > \mu_0$	$t > t_\alpha$
		$\mu \neq \mu_0$	$t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$	$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}; \sigma_1 \text{ and } \sigma_2 \text{ known}$	$\mu_1 - \mu_2 < d_0$	$z < -z_\alpha$
		$\mu_1 - \mu_2 > d_0$	$z > z_\alpha$
		$\mu_1 - \mu_2 \neq d_0$	$z < -z_{\alpha/2} \text{ or } z > z_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$	$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}}; v = n_1 + n_2 - 2, \sigma_1 = \sigma_2 \text{ but unknown, } s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$	$\mu_1 - \mu_2 < d_0$	$t < -t_\alpha$
		$\mu_1 - \mu_2 > d_0$	$t > t_\alpha$
		$\mu_1 - \mu_2 \neq d_0$	$t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$	$t' = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}; v = \frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}, \sigma_1 \neq \sigma_2 \text{ and unknown}$	$\mu_1 - \mu_2 < d_0$	$t' < -t_\alpha$
		$\mu_1 - \mu_2 > d_0$	$t' > t_\alpha$
		$\mu_1 - \mu_2 \neq d_0$	$t' < -t_{\alpha/2} \text{ or } t' > t_{\alpha/2}$
$\mu_D = d_0$ paired observations	$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}; v = n - 1$	$\mu_D < d_0$	$t < -t_\alpha$
		$\mu_D > d_0$	$t > t_\alpha$
		$\mu_D \neq d_0$	$t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$